



Unggi Lee

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Real-time systems · DNN inference offloading · Image-adaptive compression

Education

Daegu Gyeongbuk Institute of Science and Technology (DGIST)

M.S. IN COMPUTER SCIENCE

Daegu, S.Korea

Feb. 2026 - Feb. 2028 (Expected)

- Real-Time Computing Lab (RTCL)

Daegu Gyeongbuk Institute of Science and Technology (DGIST)

B.S. IN ELECTRICAL ENGINEERING AND COMPUTER SCIENCE

Daegu, S.Korea

Feb. 2020 - Feb. 2026

- Total GPA of 3.82 / 4.30

Research Experience

Real-Time Computing Lab (RTCL), DGIST

GRADUATE RESEARCH ASSISTANT

Daegu, S.Korea

Feb. 2026 - Present

- Real-time systems, with a focus on DNN inference offloading between edge devices and servers under latency constraints.
- Image-adaptive JPEG compression: per-image quality factor and quantization table optimization, so that offloaded inference holds up under variable network bandwidth.

Project Experience

WardrobeRL: Prompt-Aware Reinforcement Learning Based Top-Bottom Recommendation

Daegu, S.Korea

TEAM MEMBER, SECOND AUTHOR — REINFORCEMENT LEARNING COURSE PROJECT, PUBLISHED AT ICTC 2025

2025

- Framed prompt-conditioned outfit recommendation as sequential decision-making rather than one-shot retrieval.
- Reduced a 25M-combination action space with balanced K-means over CLIP embeddings — 5,000 tops and 5,000 bottoms from DressCode into 50 clusters of 100 items each — letting a model-free PPO agent select a cluster, then an item.
- Designed a multimodal reward fusing CLIP text-image similarity with a lightweight MLP proxy for human preference, trained on 2,500 CatVTON virtual try-on examples.
- Pre-training on the CLIP reward before fine-tuning on the human-feedback proxy started ~43% lower in loss and reached the threshold in ~8% fewer steps.
- Lifted CLIP-T text alignment from 0.833 to 0.924 over the untrained baseline, converging in under four hours on an A100.

Noise Map-Driven Dynamic Frame Processing for Improved Object Detection in Low-Light Conditions

Daegu, S.Korea

UNDERGRADUATE GROUP RESEARCH PROGRAM (UGRP), TEAM OF FOUR

2024

- Placed an enhancement front-end before YOLO — Zero-DCE for low-light enhancement, CBDNet for noise-map estimation, and MFDNet for denoising — integrated directly into the detector's data loader.
- Held the pipeline to real time by exploiting temporal redundancy between frames, processing one frame in five at 20 FPS.
- Improved mAP by roughly 1% over the unenhanced baseline on BDD100k.

Publications

WardrobeRL: Prompt-Aware Reinforcement Learning Based Top-Bottom Recommendation

ICTC 2025

MINSEOK OH, UNGGI LEE, HYUNMIN KIM, DOOSEOK LEE

Oct. 2025

- 2025 16th International Conference on Information and Communication Technology Convergence (ICTC) — [PDF](#) / [IEEE Xplore](#)

International Experience

University of California, Los Angeles (UCLA)

SUMMER SESSION

Los Angeles, CA, USA

Jun. - Aug. 2023

Skills

Programming C, C++, Python
Deep Learning PyTorch
Computer Vision OpenCV

